Aufgabe 1

1.1

Code

```sql
-- 1.1

INSERT INTO my_employee VALUES (1, 'Fawcett', 'Mattheuw', 'mface', 2895, SYSDATE - 1);
INSERT INTO my_employee VALUES (2, 'Paltrow', 'Vivien', NULL, 3860, TO_DATE('01.05.2024', 'DD.MM.YYYY'));

--1.2
INSERT INTO my_employee (id, last_name, first_name, email, salary, hire_date)
VALUES (3, 'Peck', 'Fred', 'fpeck', 1100, TO_DATE('01.01.2025', 'DD.MM.YYYY'));

INSERT INTO my_employee (id, last_name, first_name, email, salary, hire_date)
VALUES (4, 'Olivier', 'Kevin', 'kolivi', 2750, TO_DATE('15.02.2025', 'DD.MM.YYYY'));

INSERT INTO my_employee (id, last_name, first_name, email, salary, hire_date)
VALUES (5, 'Dean', 'John', 'jdean', 1550, NULL);

SELECT *
FROM my_employee;

COMMIT;
```

Ausgabe

```
1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

Commit complete.
```

	ID	LAST_NAME	FIRST_NAME	EMAIL	SALARY	HIRE_DATE
1	1	Fawcett	Mattheuw	mface	2895	09-JUN-26
2	2	Paltrow	Vivien	(null)	3860	01-MAY-24
3	3	Peck	Fred	fpeck	1100	01-JAN-25
4	4	Olivier	Kevin	kolivi	2750	15-FEB-25
5	5	Dean	John	jdean	1550	(null)

1.2

Code

```sql
-- 2 -- last_name von id 5 ändern, email ändern

UPDATE my_employee
SET last_name = 'Voight', email = 'jvoigh'
WHERE id = 5;
```

Ausgabe

```
1 row updated.
```

	ID	LAST_NAME	FIRST_NAME	EMAIL	SALARY	HIRE_DATE
1	1	Fawcett	Mattheuw	mface	2895	09-JUN-26
2	2	Paltrow	Vivien	(null)	3860	01-MAY-24
3	3	Peck	Fred	fpeck	1100	01-JAN-25
4	4	Olivier	Kevin	kolivi	2750	15-FEB-25
5	5	Voight	John	jvoigh	1550	(null)

1.3

Code

```sql
UPDATE my_employee
SET salary = GREATEST(salary * 1.1, 1250)
WHERE salary < 2000;
```

Ausgabe

```
2 rows updated.
```

	ID	LAST_NAME	FIRST_NAME	EMAIL	SALARY	HIRE_DATE
1	1	Fawcett	Mattheuw	mface	2895	09-JUN-26
2	2	Paltrow	Vivien	(null)	3860	01-MAY-24
3	3	Peck	Fred	fpeck	1250	01-JAN-25
4	4	Olivier	Kevin	kolivi	2750	15-FEB-25
5	5	Voight	John	jvoigh	1705	(null)

1.4

Code

```sql
UPDATE my_employee
SET email = LOWER(
   SUBSTR(first_name, 1, 1) || SUBSTR(last_name, 1, 5)
)
WHERE email IS NULL;
```

Ausgabe

```
1 row updated.
```

	ID	LAST_NAME	FIRST_NAME	EMAIL	SALARY	HIRE_DATE
1	1	Fawcett	Mattheuw	mface	2895	09-JUN-26
2	2	Paltrow	Vivien	vpaltr	3860	01-MAY-24
3	3	Peck	Fred	fpeck	1250	01-JAN-25
4	4	Olivier	Kevin	kolivi	2750	15-FEB-25
5	5	Voight	John	jvoigh	1705	(null)

1.5

Code

```sql
DELETE FROM my_employee
WHERE first_name = 'Fred' AND last_name = 'Peck';

COMMIT;
```

Ausgabe

	ID	LAST_NAME	FIRST_NAME	EMAIL	SALARY	HIRE_DATE
1	1	Fawcett	Mattheuw	mface	2895	09-JUN-26
2	2	Paltrow	Vivien	vpaltr	3860	01-MAY-24
3	4	Olivier	Kevin	kolivi	2750	15-FEB-25
4	5	Voight	John	jvoigh	1705	(null)

```
1 row deleted.
```

1.6a

Code

```sql
INSERT INTO my_employee (id, last_name, first_name, email, salary, hire_date)
SELECT employee_id, last_name, first_name, email, salary, hire_date
FROM employees;

SELECT *
FROM my_employee;

COMMIT;
```

Ausgabe

	ID	LAST_NAME	FIRST_NAME	EMAIL	SALARY	HIRE_DATE
1	1	Fawcett	Mattheuw	mface	2895	09-JUN-26
2	2	Paltrow	Vivien	vpaltr	3860	01-MAY-24
3	4	Olivier	Kevin	kolivi	2750	15-FEB-25
4	5	Voight	John	jvoigh	1705	(null)
5	100	King	Steven	SKING	24000	17-JUN-87
6	101	Kochhar	Neena	NKOCHHAR	17000	21-SEP-89
7	102	De Haan	Lex	LDEHAAN	17000	13-JAN-93
8	103	Hunold	Alexander	AHUNOLD	9000	03-JAN-90
9	104	Ernst	Bruce	BERNST	6000	21-MAY-91
10	107	Lorentz	Diana	DLORENTZ	4200	07-FEB-99
11	124	Mourgos	Kevin	KMOURGOS	5800	16-NOV-99
12	141	Rajs	Trenna	TRAJS	3500	17-OCT-95
13	142	Davies	Curtis	CDAVIES	3100	29-JAN-97
14	143	Matos	Randall	RMATOS	2600	15-MAR-98
15	144	Vargas	Peter	PVARGAS	2500	09-JUL-98
16	149	Zlotkey	Eleni	EZLOTKEY	10500	29-JAN-00
17	174	Abel	Ellen	EABEL	11000	11-MAY-96
18	176	Taylor	Jonathon	JTAYLOR	8600	24-MAR-98
19	178	Grant	Kimberely	KGRANT	7000	24-MAY-99
20	200	Whalen	Jennifer	JWHALEN	4400	17-SEP-87
21	201	Hartstein	Michael	MHARTSTE	13000	17-FEB-96
22	202	Fay	Pat	PFAY	6000	17-AUG-97
23	205	Higgins	Shelley	SHIGGINS	12000	07-JUN-94
24	206	Gietz	William	WGIETZ	8300	07-JUN-94

```
20 rows inserted.

Commit complete.
```

1.6b

Code

```
UPDATE my_employee
SET salary = salary + 500
WHERE id IN (
    SELECT e.id
    FROM my_employee e JOIN employees e_orig ON (e.id = e_orig.employee_id)
    WHERE e.id IN (
        SELECT employee_id
        FROM job_history
        GROUP BY employee_id
        HAVING COUNT(*) >= 2
    )
    AND e_orig.job_id NOT IN ( -- Alle jobs die e_orig.employee_id = e.id in der vergangenheit gemacht hat
        SELECT job_id
        FROM job_history jh
        WHERE e.id = jh.employee_id
        GROUP BY job_id
    )
);
```

Ausgabe

```
1 row updated.
```

	ID	LAST_NAME	FIRST_NAME	EMAIL	SALARY	HIRE_DATE
1	1	Fawcett	Mattheuw	mface	2895	09-JUN-26
2	2	Paltrow	Vivien	vpaltr	3860	01-MAY-24
3	4	Olivier	Kevin	kolivi	2750	15-FEB-25
4	5	Voight	John	jvoigh	1705	(null)
5	100	King	Steven	SKING	24000	17-JUN-87
6	101	Kochhar	Neena	NKOCHHAR	17500	21-SEP-89
7	102	De Haan	Lex	LDEHAAN	17000	13-JAN-93
8	103	Hunold	Alexander	AHUNOLD	9000	03-JAN-90
9	104	Ernst	Bruce	BERNST	6000	21-MAY-91
10	107	Lorentz	Diana	DLORENTZ	4200	07-FEB-99
11	124	Mourgos	Kevin	KMOURGOS	5800	16-NOV-99
12	141	Rajs	Trenna	TRAJS	3500	17-OCT-95
13	142	Davies	Curtis	CDAVIES	3100	29-JAN-97
14	143	Matos	Randall	RMATOS	2600	15-MAR-98
15	144	Vargas	Peter	PVARGAS	2500	09-JUL-98
16	149	Zlotkey	Eleni	EZLOTKEY	10500	29-JAN-00
17	174	Abel	Ellen	EABEL	11000	11-MAY-96
18	176	Taylor	Jonathon	JTAYLOR	8600	24-MAR-98
19	178	Grant	Kimberely	KGRANT	7000	24-MAY-99
20	200	Whalen	Jennifer	JWHALEN	4400	17-SEP-87
21	201	Hartstein	Michael	MHARTSTE	13000	17-FEB-96
22	202	Fay	Pat	PFAY	6000	17-AUG-97
23	205	Higgins	Shelley	SHIGGINS	12000	07-JUN-94
24	206	Gietz	William	WGIETZ	8300	07-JUN-94

1.7

Code

```sql
DELETE FROM my_employee
WHERE hire_date <= (
   SELECT MAX(hire_date)
   FROM my_employee
   WHERE last_name = 'Hunold' OR last_name = 'De Haan'
);
```

Ausgabe

```
6 rows deleted.
```

	ID	LAST_NAME	FIRST_NAME	EMAIL	SALARY	HIRE_DATE
1	1	Fawcett	Mattheuw	mface	2895	09-JUN-26
2	2	Paltrow	Vivien	vpaltr	3860	01-MAY-24
3	4	Olivier	Kevin	kolivi	2750	15-FEB-25
4	5	Voight	John	jvoigh	1705	(null)
5	107	Lorentz	Diana	DLORENTZ	4200	07-FEB-99
6	124	Mourgos	Kevin	KMOURGOS	5800	16-NOV-99
7	141	Rajs	Trenna	TRAJS	3500	17-OCT-95
8	142	Davies	Curtis	CDAVIES	3100	29-JAN-97
9	143	Matos	Randall	RMATOS	2600	15-MAR-98
10	144	Vargas	Peter	PVARGAS	2500	09-JUL-98
11	149	Zlotkey	Eleni	EZLOTKEY	10500	29-JAN-00
12	174	Abel	Ellen	EABEL	11000	11-MAY-96
13	176	Taylor	Jonathon	JTAYLOR	8600	24-MAR-98
14	178	Grant	Kimberely	KGRANT	7000	24-MAY-99
15	201	Hartstein	Michael	MHARTSTE	13000	17-FEB-96
16	202	Fay	Pat	PFAY	6000	17-AUG-97
17	205	Higgins	Shelley	SHIGGINS	12000	07-JUN-94
18	206	Gietz	William	WGIETZ	8300	07-JUN-94

Aufgabe 2

2.1

Code

```
MERGE INTO bonus b
USING employees e
   ON (e.employee_id = b.employee_id)
WHEN MATCHED THEN   -- Mitarbeiter existiert in Bonus
                    -- department_id = 80 werden gelöscht
                    -- über 11000€ Leute werden gelöscht
                    -- Erhöhung um 15% des aktuellen Bonusbetrages
   UPDATE SET b.bonus = b.bonus * 1.15
   DELETE WHERE (e.salary > 11000) OR (department_id = 80)

WHEN NOT MATCHED THEN -- Mitarbeiter hat noch keinen bonus erhalten
                      -- Bonus beträgt 1 % des Gehalts (e.salary)
                      -- department_id = 80 werden nicht eingefügt
                      -- Gehalt über 11000 wird nicht eingefügt
                      -- Null soll einen Bonus bekommen können
   INSERT (employee_id, bonus)
   VALUES (e.employee_id, e.salary * 0.01)
   WHERE ((e.department_id != 80) OR (e.department_id IS NULL)) AND (e.salary <= 11000);
```

Ausgabe

```
15 rows merged.
```

	EMPLOYEE_ID	BONUS
1	124	115
2	141	115
3	142	115
4	143	115
5	144	115
6	103	90
7	104	60
8	107	42
9	178	70
10	200	44
11	202	60
12	206	83

Aufgabe 3

3.1

Code

```
-- 1
CREATE OR REPLACE VIEW schulung (Personalnummer, Vorname, Nachname, Abteilung, Bundesland) AS
SELECT employee_id, first_name, last_name, department_name, state_province
FROM employees
  JOIN departments USING (department_id)
  JOIN locations USING (location_id);

SELECT * FROM schulung;
```

Ausgabe

View SCHULUNG created.

	PERSONALNUMMER	VORNAME	NACHNAME	ABTEILUNG	BUNDESLAND
1	100	Steven	King	Executive	Washington
2	101	Neena	Kochhar	Executive	Washington
3	102	Lex	De Haan	Executive	Washington
4	103	Alexander	Hunold	IT	Texas
5	104	Bruce	Ernst	IT	Texas
6	107	Diana	Lorentz	IT	Texas
7	124	Kevin	Mourgos	Shipping	California
8	141	Trenna	Rajs	Shipping	California
9	142	Curtis	Davies	Shipping	California
10	143	Randall	Matos	Shipping	California
11	144	Peter	Vargas	Shipping	California
12	149	Eleni	Zlotkey	Sales	Oxford
13	174	Ellen	Abel	Sales	Oxford
14	176	Jonathon	Taylor	Sales	Oxford
15	200	Jennifer	Whalen	Administration	Washington
16	201	Michael	Hartstein	Marketing	Ontario
17	202	Pat	Fay	Marketing	Ontario
18	205	Shelley	Higgins	Accounting	Washington
19	206	William	Gietz	Accounting	Washington

3.2

Code

```
CREATE OR REPLACE VIEW DEPT90 (EMPNO, EMPLOYEE, DEPTNO) AS
SELECT employee_id, last_name, department_id
FROM employees
WHERE department_id = 90
WITH CHECK OPTION CONSTRAINT dept90_department_id_check;

SELECT * FROM DEPT90;
```

Ausgabe

View DEPT90 created.

	EMPNO	EMPLOYEE	DEPTNO
1	100	King	90
2	101	Kochhar	90
3	102	De Haan	90

3.3a

Code

```
UPDATE dept90
SET deptNo = 60
WHERE employee = 'Kochhar';
```

Ausgabe

```
Error starting at line : 1 in command -
UPDATE dept90
SET deptNo = 60
WHERE employee = 'Kochhar'
Error at Command Line : 1 Column : 8
Error report -
SQL Error: ORA-01402: view WITH CHECK OPTION where-clause violation
```

3.3b

Code

```
UPDATE dept90
SET employee = 'The King'
WHERE empNo = 100;
```

Ausgabe

```
1 row updated.
```

	EMPNO	EMPLOYEE	DEPTNO
1	100	The King	90
2	101	Kochhar	90
3	102	De Haan	90

3.3c

Warum funktioniert a) nicht und b) schon?
- Durch die Änderung der Abteilungsnummer auf 60 würde der Datensatz aus der View rausfallen. Genau das verhindert die check-option.
- Durch die Änderung von 'King' auf 'The King' fällt dieser Datensatz nicht aus der Sicht - daher wird das statement ausgeführt.
Durch ein 'ROLLBACK' wurde die Änderung rückgängig gemacht.

	EMPNO	EMPLOYEE	DEPTNO
1	100	King	90
2	101	Kochhar	90
3	102	De Haan	90

3.4

Code

```
CREATE OR REPLACE VIEW annual_salary (employee, employee_id, annual_salary) AS
SELECT first_name || ' ' || last_name, employee_id, salary * 14
FROM employees;

SELECT * FROM annual_salary;
```

Ausgabe

```
View ANNUAL_SALARY created.
```

	EMPLOYEE	EMPLOYEE_ID	ANNUAL_SALARY
1	Steven King	100	336000
2	Neena Kochhar	101	238000
3	Lex De Haan	102	238000
4	Alexander Hunold	103	126000
5	Bruce Ernst	104	84000
6	Diana Lorentz	107	58800
7	Kevin Mourgos	124	81200
8	Trenna Rajs	141	49000
9	Curtis Davies	142	43400
10	Randall Matos	143	36400
11	Peter Vargas	144	35000
12	Eleni Zlotkey	149	147000
13	Ellen Abel	174	154000
14	Jonathon Taylor	176	120400
15	Kimberely Grant	178	98000
16	Jennifer Whalen	200	61600
17	Michael Hartstein	201	182000
18	Pat Fay	202	84000
19	Shelley Higgins	205	168000
20	William Gietz	206	116200

3.4 DMLs testen bei Angestellten 100 und 178

Code

```
UPDATE annual_salary
SET employee = 'Stephan König'
WHERE employee_id = 100;

UPDATE annual_salary
SET annual_salary = 0
WHERE employee_id = 100;

UPDATE annual_salary
SET employee_id = 50
WHERE employee_id = 178;
```

Ausgabe

```
Error starting at line : 1 in command -
UPDATE annual_salary
SET annual_salary = 0
WHERE employee_id = 100
Error at Command Line : 2 Column : 5
Error report -
SQL Error: ORA-01733: virtual column not allowed here
```

```
Error starting at line : 1 in command -
UPDATE annual_salary
SET employee = 'Stephan König'
WHERE employee_id = 100
Error at Command Line : 2 Column : 5
Error report -
SQL Error: ORA-01733: virtual column not allowed here
```

14	Jonathon Taylor	176	120400
15	Kimberely Grant	50	98000
16	Jennifer Whalen	200	61600

Note: Das erste statement kann nicht funktionieren, weil 'employee' aus mehreren Attributen zusammengebaut ist. Es wäre nicht klar, was genau in der Basistabelle geändert werden muss.
Auch das annual_salary anzupassen funktioniert nicht, weil im Select auf salary*14 projeziert wird.
Die employee_id wird direkt aus der Haupttabelle "weitergegeben" und ist daher änderbar, wenn kein Fremdschlüssel auf ihn referenziert.

Note: employee 178 ist jetzt employee 50

Code

```sql
-- delete
DELETE FROM annual_salary
WHERE employee_id = 100;

DELETE FROM annual_salary
WHERE employee_id = 50;
```

Ausgabe

```
Error starting at line : 1 in command -
DELETE FROM annual_salary
WHERE employee_id = 100
Error report -
ORA-02292: integrity constraint (S2510307051.DEPT_MGR_FK) violated - child record found

https://docs.oracle.com/error-help/db/ora-02292/

More Details :
https://docs.oracle.com/error-help/db/ora-02292/

1 row deleted.
```

Note: Datensatz mit der id 100 zu löschen funktioniert nicht, weil auf diesen als Fremdschlüssel referenziert wird (z.B. steht er bei jemand anderen als manager)
Bei Datensatz mit der id 50 (vorher 178) ist das nicht der Fall, deshalb lässt er sich löschen.

Aufgabe 4

4.1

Code

```
SELECT e.*
FROM employees e
   JOIN employees m ON (e.manager_id = m.employee_id)
   WHERE e.hire_date < m.hire_date;
```

Ausgabe

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	200	Jennifer	Whalen	JWHALEN	515.123.4444	17-SEP-87	AD_ASST	4400	(null)	101	10
2	103	Alexander	Hunold	AHUNOLD	590.423.4567	03-JAN-90	IT_PROG	9000	(null)	102	60
3	142	Curtis	Davies	CDAVIES	650.121.2994	29-JAN-97	ST_CLERK	3100	(null)	124	50
4	143	Randall	Matos	RMATOS	650.121.2874	15-MAR-98	ST_CLERK	2600	(null)	124	50
5	144	Peter	Vargas	PVARGAS	650.121.2004	09-JUL-98	ST_CLERK	2500	(null)	124	50
6	141	Trenna	Rajs	TRAJS	650.121.8009	17-OCT-95	ST_CLERK	3500	(null)	124	50
7	178	Kimberely	Grant	KGRANT	011.44.1644.429263	24-MAY-99	SA_REP	7000	0.15	149	(null)
8	174	Ellen	Abel	EABEL	011.44.1644.429267	11-MAY-96	SA_REP	11000	0.3	149	80
9	176	Jonathon	Taylor	JTAYLOR	011.44.1644.429265	24-MAR-98	SA_REP	8600	0.2	149	80

4.2

Code

```
SELECT e.last_name, e.first_name, e.job_id, d.department_name
FROM employees e
   LEFT JOIN departments d ON (e.department_id = d.department_id)
   -- employee bleibt erhalten, wenn department_id Null ist
   JOIN (
      SELECT job_id, AVG(salary) AS average_salary
      FROM employees
      GROUP BY job_id
   ) avg_sal ON (e.job_id = avg_sal.job_id)
WHERE e.salary > avg_sal.average_salary;
```

Ausgabe

	LAST_NAME	FIRST_NAME	JOB_ID	DEPARTMENT_NAME
1	Rajs	Trenna	ST_CLERK	Shipping
2	Davies	Curtis	ST_CLERK	Shipping
3	Hunold	Alexander	IT_PROG	IT
4	Abel	Ellen	SA_REP	Sales